A REPRESENTABLE MORPHISM IS UNIVERSALLY SUBMERSIVE EXACTLY WHEN ALL ITS BASE CHANGES ARE SUBMERSIVE ON SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The definition of a universally submersive representable morphism of algebraic stacks, tested after base change by schemes. Referred to in the statement (source lines 2570-2571) as a pointer to the chapter on properties of stacks. Formulates the two conditions being compared.
- The description of the topological space of an algebraic stack through a smooth cover, with the induced maps open and surjective and the induced map to the fibre product of the spaces surjective. Invoked inside the counted proof (source lines 2583-2597) as a pointer to the same chapter. Descends submersiveness from a smooth cover to the base change of the stack.

Source reference. Original reference stacks-properties-section-points: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2488–2488. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-section-points, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-section-properties-morphisms: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 248–248. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-section-properties-morphisms, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks which is representable by algebraic spaces. The following are equivalent*

(1) *f is universally submersive (as in Properties of Stacks, Section stacks-properties-section-properties-morphisms), and*
(2) *for every morphism of algebraic stacks $\mathcal{Z} \to \mathcal{Y}$ the morphism of topological spaces $|\mathcal{Z} \times_{\mathcal{Y}} \mathcal{X}| \to |\mathcal{Z}|$ is submersive.*

Proof. Assume (1), and let $\mathcal{Z} \to \mathcal{Y}$ be as in (2). Choose a scheme V and a surjective smooth morphism $V \to \mathcal{Z}$. By assumption the morphism $V \times_{\mathcal{Y}} \mathcal{X} \to V$ of algebraic spaces is universally submersive, in particular the map $|V \times_{\mathcal{Y}} \mathcal{X}| \to |V|$ is submersive. By Properties of Stacks, Section stacks-properties-section-points in the commutative diagram

$$\begin{array}{ccc} |V \times_{\mathcal{Y}} \mathcal{X}| & \longrightarrow & |\mathcal{Z} \times_{\mathcal{Y}} \mathcal{X}| \\ \downarrow & & \downarrow \\ |V| & \longrightarrow & |\mathcal{Z}| \end{array}$$

the horizontal arrows are open and surjective, and moreover

$$|V \times_{\mathcal{Y}} \mathcal{X}| \longrightarrow |V| \times_{|\mathcal{Z}|} |\mathcal{Z} \times_{\mathcal{Y}} \mathcal{X}|$$

is surjective. Hence as the left vertical arrow is submersive it follows that the right vertical arrow is submersive. This proves (2). The implication (2) $\Rightarrow$ (1) follows from the definitions. $\square$